The Spectral Admissibility Sub-Programme

Presentation Note 1

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Abstract

The spectral admissibility sub-programme of the Cosmochrony corpus addresses a single central question: which spectral sectors of the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ survive the Born–Infeld bounded-flux constraint? The answer organises the entire O-series (papers O1–O32 and five precursor papers) into a logically closed derivation chain:

$$\text{bounded flux} \implies \text{admissibility envelope} \implies \text{Weil-sector capacity} \implies \delta_{\text{pair}} \implies \beta^* \approx 0.126.$$

This note maps the constituent papers, identifies four internal phases, records the status of every result as proved, structural, or open, and states the two remaining open deliverables: the pointwise analytical proof of hypothesis [H-color] (partially resolved at three levels in O32) and the extension of the numerical campaign to $q = 401$.

Keywords. spectral admissibility; Born–Infeld constraint; Weil representation; Heisenberg group; pair observable; capacity exponent; lepton mass hierarchy; SU(3); colour triplet; non-injective projection; presentation note.

Contents

1 Motivation and Central Question

The spectral admissibility sub-programme answers the following question.

Central question. The non-injective projection Π of the Cosmochrony framework acts on the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, decomposed into irreducible blocks V_c indexed by characters $c \in (\mathbb{Z}/q\mathbb{Z})^\times$. The Born–Infeld saturation constraint bounds the projective flux carried by each mode: $A_n \le A_n^{\max} := c_\chi/\sqrt{\lambda_n}$, where λ_n is the Laplacian eigenvalue at BFS depth n and c_χ is the BI saturation constant. *Which combinations of Weil sectors remain admissible under this constraint, and what capacity exponent δ_{pair} do they carry?*

The practical importance of this question is that δ_{pair} governs the cascade exponent β^* via the structural relation $\beta^* \approx 1/(\delta_{\mathrm{pair}} + \frac{1}{2})$ established in the O-series. The value $\beta^* \approx 0.126$ matches the phenomenological window $\beta^* \in (0.09, 0.13)$ derived from the charged-lepton mass ratios $m_e : m_\mu : m_\tau$, making this sub-programme the primary quantitative connection between the projective framework and Standard Model observables.

The sub-programme is also the hardest to navigate: it spans over thirty papers written over a period of successive obstruction identifications and resolutions. This note is intended as a structured entry point, not a summary of results.

2 Position in the Cosmochrony Programme

The Cosmochrony corpus is organised into three branches. Branch I establishes the axiomatic primitive: four axioms (A1–A4) suffice to derive $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and its Weil representation as theorems [4, 15, 14]. Branch II is the spectral admissibility sub-programme (this note). Branch III derives quantum mechanics, spacetime geometry, gauge structure, and fermionic matter from Branch I axioms and Branch II spectral data.

The spectral admissibility sub-programme therefore occupies a pivotal position: it is the *computational engine* of the corpus. It takes the algebraic output of Branch I (the Weil representation and its decomposition into irreducible blocks) and produces the two inputs that Branch III depends on:

1. the identification of the admissible sector as the spin-$\frac{1}{2}$ representation of $\mathrm{SU}(2)$ embedded via the admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$;
2. the numerical value $\beta^* \approx 0.126$ and its representation-theoretic interpretation as the scaling exponent of Hilbert–Schmidt norm growth along trajectories in the minimal admissible non-abelian sector $\mathfrak{su}(2)$.

Remark 1 (Vocabulary note)**.** Throughout this note, the term *admissibility constraint* replaces the earlier "relaxation" vocabulary of the preliminary Cosmochrony formulation. The substrate χ is static: no substrate dynamics or relaxation mechanism is postulated at the foundational level. Observable evolution is an induced ordering of admissible projections, not a dynamics of χ itself.

3 Logical Chain of the Sub-Programme

The sub-programme is organised around the following derivation chain.

$$\underbrace{c_\chi}_{\text{BI saturation}} \implies \underbrace{A_n^{\max} = c_\chi/\sqrt{\lambda_n}}_{\text{admissibility envelope}} \implies \underbrace{\sigma_{\mathrm{pair}}^{\mathrm{can}}(n)}_{\text{pair capacity}} \implies \underbrace{\delta_{\mathrm{pair}}}_{\text{capacity exponent}} \implies \underbrace{\beta^* \approx 0.126}_{\text{cascade exponent}} . \tag{1}$$

Each arrow in (1) corresponds to a distinct analytical or numerical sub-problem, resolved at a different stage of the sub-programme. The transfer mechanism $c_\chi \to \delta_{\text{pair}} \to \beta^*$ is unconditional at the level of the admissible fibre structure (O24, [36]): no free parameter is adjusted to produce $\beta^* \approx 0.126$. The asymptotic numerical stabilisation of $\delta_{\text{pair}}(q)$ as $q \to \infty$ is a separate analytical and numerical question, recorded as open in Section 7.4. The sole conditionality on the *group-theoretic* side concerns the SU(3) sector, which requires hypothesis [H-color] (Section 8).

Three structural facts underlie the chain and should not be confused.

1. The *parity involution* $c \leftrightarrow q - c$, derived from Born–Infeld evenness in O18 [12], is the structural origin of the pair observable $\sigma_{\text{pair}}^{\text{can}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$. The factor-of-two relation $\delta_{\text{pair}} = 2\delta_c$ is a consequence of this fibre pairing, not a normalisation artefact.

2. The *observable hierarchy*: the pair capacity $\sigma_c(n)$ (normalised BFS span in the Weil block V_c) is strictly finer than the Markov spectrum of the Weil operators. Spectral equality of characters does not imply capacity profile equality. This distinction is essential for hypothesis [H-color].

3. The *admissibility thread* $Q_8 \subset 2I \subset \mathrm{SU}(2)$ is an output of the sub-programme, not an input. Q_8 is the minimal non-abelian finite subgroup of SU(2) compatible with bounded-flux admissibility; $2I$ is the maximal admissible finite subgroup; the LPS family provides the Ramanujan approximation to SU(2) in the large-q limit.

4 Constituent Papers

The sub-programme comprises five precursor papers and thirty-two O-series papers. Table 1 provides an overview organised by internal phase.

4.1 Precursor papers: SU(2) admissibility sub-programme

Four papers precede the O-series proper and explore spectral admissibility on Cayley graphs of finite SU(2) subgroups.

SpectralAdmissibility [29] carries out the spectral admissibility programme on the chain $Q_8 \subset 2I \subset \mathrm{PSL}(2, \mathbb{F}_q)$. It derives the admissibility hierarchy $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ in the linearised regime and shows the non-linear DBI correction preserves it. For the SU(3) chain $\Delta(27) \subset \Sigma(168)$, it establishes the co-admissibility of the fundamental and anti-fundamental representations via character theory.

SpectralCapacity [30] introduces the capacity functional $\mathcal{C}_\alpha(G, S)$ and proves the binary-polyhedral maximality conjecture: $2I$ uniquely maximises $\mathcal{C}_\alpha(G, S)$ for all $\alpha > 0$ among finite SU(2) subgroups with neutral generating sets. The proof is closed analytically for degrees $d \in \{6, 12, 24\}$ (Theorem 8.1 of that paper).

SpectralGramRigidity [18] shows that the geometry of neutral generating sets on SU(2) is fully encoded in pairwise character values $G_{st} = -\frac{1}{2}\chi_\rho(st)$.

ThreeGenerations [34] derives the three-generation structure from the ADE stratigraphic levels of binary Cayley graphs (structural).

SpectralRelaxation [6] establishes $\Lambda_{\text{proj}} \asymp \lambda_2$ for LPS expander families, providing the entry point for the O-series dynamics.

4.2 O-series core: LPS phase (O1–O8)

The opening O-series papers work on Lubotzky–Phillips–Sarnak expander graphs, which have exponential shell growth $|B_n| \sim p^n$ and therefore represent the most natural large-group setting for bounded-flux dynamics.

O1 [24] shows that LPS saturation is a fixed-valence artefact; growing valence restores the pre-saturation window.

O3 [22] derives the phenomenological target window $\beta^* \in (0.09, 0.13)$ from the charged-lepton mass ratios $m_e : m_\mu : m_\tau$ (structural).

O4 [32] proves the structural upper bound $\beta \leq 1$ from the BI flux constraint and the Cheeger isoperimetric inequality.

O5 [3] establishes the obstruction theorem: the β gap cannot be closed by any purely representation-theoretic or fixed-representation approach.

O6 [11] proves matrix-level confinement of the cascade exponent.

O7 [20] reformulates the target as a capacity exponent $\delta \in [7.4, 10.6]$ on three-step path fingerprints (structural).

O8 [1] establishes the geometric obstruction closing the LPS phase: exponential shell growth compresses the entire pre-saturation window into $O(\log q)$ BFS steps, preventing stable slope estimation.

4.3 O-series core: Heisenberg transition (O9–O15)

The geometric obstruction of O8 forces a change of arena. The Cayley graph $\mathrm{Cay}(\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$ has polynomial ball growth $|B_n| \sim Cn^4$ (Bass–Guivarc'h theorem, homogeneous dimension $D = 4$), restoring an exploitable window of $\Theta(q)$ BFS steps.

O9 [21] resolves the geometric obstruction by switching to Heisenberg graphs. Polynomial growth is confirmed; the window depth scales as $\Theta(q)$.

O10 [19] performs the first large-q computation ($q = 101$, $|G_q| = 1{,}030{,}301$) and identifies an algorithmic obstruction: the dense TensorSketch fingerprint is structurally mismatched to the problem, requiring sketch dimension $D \gg q^4$.

O11 [37] resolves the algorithmic obstruction via a bidimensional Fourier-character proxy for the Weil-block decomposition; a stable estimate $\hat{\delta}_{\mathrm{cap}} \approx 3.4$ is extracted at $q = 307$.

O12 [8] implements the exact Weil projection, closes the representation-level ambiguity left by O11, and extracts $\hat{\delta}_{\mathrm{exact}} \approx 4.3$–$4.8$ for $q \in \{29, 53, 61\}$ ($R^2 > 0.98$).

O13 [7] confirms asymptotic stability of the exact Weil-block capacity over the extended prime range and establishes monotone decrease of $\hat{\delta}_{\mathrm{exact}}(q)$, definitively refuting the finite-size hypothesis.

O14 [17] corrects the $\delta \mapsto \beta^*$ relation via block normalisation and central-phase contributions.

O15 [16] establishes the block-level no-go: no reweighting of individual block data recovers the admissible window $\delta \in [7.4, 10.6]$ (proved, Corollary 4.8 of that paper). This result forces the introduction of the pair observable in O16.

4.4 Pair observable and transfer chain (O16–O24)

The block-level no-go of O15 is structurally informative: it identifies the wrong observable class, not a defect of the transfer mechanism. The physically correct observable is the *pair capacity* $\sigma^{\mathrm{can}}_{\mathrm{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ on conjugate pairs $\{c, q - c\}$.

O16 [9] identifies the observable mismatch. The exponent doubling $\delta_{\mathrm{pair}} = 2\delta_c \approx 7.44$ resolves the gap (structural).

O17 [10] proves that $\rho_{q-c} = \bar{\rho}_c$ is forced by fibre-level admissibility; the pair structure is derived, not postulated.

O18 [12] gives the foundational derivation of the parity involution $c \leftrightarrow q - c$. Born–Infeld parity is the only global symmetry of the BI action, ensuring each fibre of Π is exactly $\{\chi, -\chi\}$ (minimal fibre condition, proved, Theorem 4.1).

O19 [2] proves that the residual amplitude factor $r(c, q) \in \{1, 2, 3, 4\}$ does not affect δ_{pair}; the observable is pipeline-independent.

O20 [23] establishes a dynamical selection criterion for admissible modes via projective persistence (structural).

O21 [28] defines the observable saturation rank n_3^{obs} via the threshold condition $\Sigma_c(n_3) = 3$, constructs the fibre-level transfer $\delta_{\mathrm{pair}} \to \beta^*$, and resolves the O15 obstruction by identifying it as an observable-class mismatch.

O22 [27] proves that Born–Infeld projection locking forces n_3^{obs} onto a BFS shell of $\mathrm{Cay}(G_q, S_q)$: shell-alignment is a theorem, not an independent geometric hypothesis.

O23 [35] closes the complementary question: the threshold value $\Sigma_c(n_3) = 3$ is not an arithmetic input but a structural consequence of Born–Infeld fibre admissibility. The value three is the unique dimension of the imaginary subspace $\operatorname{Im}\mathbb{H}$ of the minimal admissible associative non-commutative algebra.

O24 [36] is the capstone of the analytical O-series. The Verticality Lemma ("$\ker\Pi$ is free, $\operatorname{Im}\Pi$ is constrained") closes the unconditional transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$ (Corollary 4.2: $\beta^* \approx 0.126$, proved).

4.5 Numerical campaign and closure papers (O25–O32)

With the analytical chain closed by O24, the post-O24 programme turns to three tasks: numerical validation of δ_{pair}, representation-theoretic identification of the admissible sector, and the SU(3) colour sector.

O25 [33] carries out the systematic pair-level numerical campaign across all $(q-1)/2$ conjugate pairs with $M = 50$ independent block samples per pair, for $q \in \{29, 61, 101, 151, 211\}$. Results: $\bar{\delta}_{\mathrm{pair}} \in \{8.89, 9.98, 9.55, 9.13, 8.78\}$ with inter-pair standard deviation decreasing from 0.54 to 0.11. After the O14 normalisation correction, values $\delta_{\mathrm{corr}}(q) \in [7.7, 9.0]$ fall consistently within the admissible window $[7.4, 10.6]$ for $q \in \{61, 101, 151, 211\}$.

O26 [25] establishes the quadratic completion of admissible spectral pairs via binary-icosahedral representation, providing the Level III identification criterion for the spin-$\frac{1}{2}$ sector.

O27 [26] proves that every admissible morphism $\Phi_{q,\rho}$ necessarily factors through $\mathfrak{su}(2)$ (quaternionic rigidity theorem, proved unconditionally). The identification chain becomes:

$$\text{emergence} \Rightarrow \mathfrak{su}(2)\ [\text{O27}] \Rightarrow \text{pair data} \in \mathrm{Sym}(V_\rho, \mathbb{C})\ [\text{O29}] \Rightarrow r_{\mathrm{eff}} = 3 \Rightarrow d_\rho = 2\ [\text{O29}].$$

O28 [5] confirms asymptotic calibration of the BFS window: $\delta_{\mathrm{corr}}(q) \in [7.4, 10.6]$ for all $q \in \{29, 61, 101, 151, 211\}$; the covariance C_c in $\mathrm{End}(H_{\mathrm{eff}})$ has rank exactly 3 for every conjugate pair at $q \in \{61, 101, 151\}$, with invariant eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$.

O29 [31] identifies the spin-$\frac{1}{2}$ irreducible sector $V_\rho \subset H_{\mathrm{eff}} = \mathbb{C}^3$ via the symmetric rank formula $r_{\mathrm{eff}} = d_\rho(d_\rho + 1)/2$. The unique solution $d_\rho = 2$ is confirmed numerically for $q \in \{61, 101, 151\}$.

O30 [38] derives analytically the invariant eigenvalue ratio $\lambda_1/\lambda_2 = 2$ of the per-pair covariance: Born–Infeld parity anti-linearity forces an equatorial constraint on the admissible fibre, and the factor of two is purely structural (squared symmetric-tensor normalisation of the Carnot-degree-2 central generator $Z = [X, Y]$, proved).

O31 [40] develops the structural framework for SU(3) emergence. Metaplectic intertwining of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ forces triplet co-admissibility unconditionally for the colour-adapted Cayley graph $\mathrm{Cay}(G_q, S_q^{(3)})$. SU(3) is identified as the symmetry group of the co-admissible colour triplet, proved for the standard generating set via Proposition 4.23 of O31 (single-frequency fingerprint structure). O31 also closes the spectral approach to [H-color]: exact generator-twist isospectrality does not imply colour-triplet spectral degeneracy.

O32 [39] provides the numerical test of [H-color] on the standard graph. Results for $q \in \{61, 151, 211\}$ (test, $q \equiv 1 \pmod 3$) and $q \in \{29, 101\}$ (control, $q \equiv 2 \pmod 3$): hypothesis [H-color] passes all three test primes; $q = 307$ confirms ($R_{\mathrm{var}} = 0.7$, all 4 triplets).

Table 1: Summary of the O-series by internal phase. Status codes: P = proved; S = structural; N = numerical; O = open.

Phase	Papers	Central output	Statu
Precursors	SpAdm, SpCap, SpGram, 3Gen	$Q_8 \subset 2I$, binary maximality	P/S
LPS phase	SpRel, O1–O8	$\beta \leq 1$; geometric obstruction	P
Heisenberg transition	O9–O15	Exact $\hat{\delta}_{\text{exact}} \approx 4.5$; no-go	P
Pair + transfer	O16–O24	$c_\chi \to \delta_{\text{pair}} \to \beta^*$, unconditional	P
Numerical + sector	O25–O30	$\delta_{\text{corr}} \in [7.4, 10.6]$; $d_\rho = 2$	P/N
SU(3) / colour	O31–O32	SU(3) unconditional; $[\text{H-color}]_{\text{pointwise}}$ proved	P

5 Inputs from Upstream Results

The spectral admissibility sub-programme takes the following inputs from Branch I.

1. **Weil representation.** The fact that $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and its Weil representation V_ρ are theorems of the axioms A1–A4, not additional postulates. This is established in Foundation [4] and HeisenbergStructure [14].

2. **BI saturation functional.** The Born–Infeld action with saturation constant c_χ is the unique analytic completion compatible with bounded projective transport capacity. The sub-programme takes c_χ as a parameter; its structural origin is in Branch I.

3. **Non-injectivity no-go.** The ENI paper [15] establishes that non-injectivity is a structural necessity of genuine emergence, not an interpretive option. The pair structure and the parity involution $c \leftrightarrow q - c$ are downstream consequences.

4. **Noscale principle.** The absence of an external dimensional scale at the admissibility level [13] ensures that the only dimensionful parameter available to the sub-programme is c_χ, enforcing the internal consistency of the capacity law.

No result from Branch III is an input to the sub-programme. The admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$, the value δ_{pair}, and the cascade exponent β^* are all outputs.

6 Outputs to Downstream Branches

Table 2 lists the principal outputs of the sub-programme and their downstream consumers.

7 Status of Results

7.1 Proved results

The following results are established as theorems within their stated assumptions.

1. *Binary-polyhedral maximality* (SpectralCapacity [30]): $2I$ uniquely maximises $\mathcal{C}_\alpha(G, S)$ for all $\alpha > 0$ among finite SU(2) subgroups with neutral generating sets. Closed analytically for $d \in \{6, 12, 24\}$ and by Proposition 8.11 for the dicyclic exclusion.

2. *Gram rigidity* (SpectralGramRigidity [18]): the geometry of neutral generating sets on SU(2) is fully encoded in pairwise character values.

Table 2: Principal outputs of the spectral admissibility sub-programme and their consumers.

Output	Consumer	Status
$\beta^* \approx 0.126$	SMchargemass, lepton hierarchy	P
Admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$	Q1–Q3 (quantum sector)	P
Spin-$\frac{1}{2}$ sector $V_\rho \cong \mathbb{C}^2$	Q7–Q11 (geometry)	P
$\Sigma_c(n_3) = 3$ and $\operatorname{Im}\mathbb{H} \cong \mathfrak{su}(2)$	Q7 (spatial identification)	P
SU(3) co-admissible triplet (conditional)	Q6a, Gauge note	S/O
$\delta_{\mathrm{corr}}(q) \in [7.4, 10.6]$ for $q \in \{61, \ldots, 211\}$	Open: $q \to \infty$ limit	N

3. *$\beta \leq 1$ upper bound* (O4 [32]): the structural upper bound from the BI flux constraint and the Cheeger inequality.
4. *Block-level no-go* (O15 [16]): no reweighting of individual block data recovers $\delta \in [7.4, 10.6]$; the pair observable is necessary.
5. *Parity involution* (O18 [12]): BI parity is the unique global symmetry of the BI action; the minimal fibre condition $\Pi^{-1}(x) = \{\chi, -\chi\}$ is forced.
6. *Pipeline independence of* δ_{pair} (O19 [2]): the residual amplitude factor $r(c, q)$ does not affect the pair exponent.
7. *Shell-alignment theorem* (O22 [27]): Born–Infeld projection locking forces $n_3^{\mathrm{obs}} \in \mathbb{N}$ as the first element of $L_{\mathrm{BI}} \cap N_q$.
8. *Quaternionic minimality* (O23 [35]): the threshold $\Sigma_c(n_3) = 3$ follows from the fact that $\mathbb{H}$ is the minimal admissible associative non-commutative algebra, and $\dim_{\mathbb{R}} \operatorname{Im}\mathbb{H} = 3$.
9. *Unconditional transfer chain* (O24 [36]): the Verticality Lemma closes $c_\chi \to \delta_{\mathrm{pair}} \to \beta^* \approx 0.126$ without residual conditionality on the fibre structure.
10. *Quaternionic rigidity* (O27 [26]): every admissible morphism $\Phi_{q,\rho}$ factors through $\mathfrak{su}(2)$.
11. *Eigenvalue ratio* (O30 [38]): the invariant ratio $\lambda_1/\lambda_2 = 2$ of the per-pair covariance is analytically derived from the equatorial admissible fibre constraint.
12. *Generator-twist isospectrality* (O31 [40]): the spectra of $\rho_c(P_{S_q})$ and $\rho_c(P_{\varphi_\omega(S_q)})$ are equal for all c, but this does not imply colour-triplet spectral degeneracy. This closes the spectral route to [H-color].

7.2 Structural results

The following results are derived from structural arguments that are internally consistent but not yet fully formalised as theorems.

1. *Phenomenological window* (O3 [22]): $\beta^* \in (0.09, 0.13)$ from $m_e : m_\mu : m_\tau$.
2. *Capacity exponent target* (O7 [20]): the target $\delta \in [7.4, 10.6]$ on three-step path fingerprints.

3. *Observable-class identification* (O16 [9]): the pair observable resolves the gap by exponent doubling $\delta_{\text{pair}} = 2\delta_c$.

4. *SU(3) emergence framework* (O31 [40]): the metaplectic intertwining of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ forces triplet co-admissibility unconditionally for the colour-adapted Cayley graph.

7.3 Numerical evidence

The following results are numerically established but not yet derived analytically.

1. $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ (O25 [33]): for $q \in \{61, 101, 151, 211\}$ after O14 normalisation; consistently within the admissible window $[7.4, 10.6]$.

2. *Spin-$\frac{1}{2}$ sector* (O29 [31]): $r_{\text{eff}} = 3$ and $d_\rho = 2$ confirmed for $q \in \{61, 101, 151\}$.

3. *Eigenvalue structure* $[1 : \frac{1}{2} : \frac{1}{2}]$ (O28 [5]): robust across all conjugate pairs and all tested primes.

4. [H-color] *numerical support* (O32 [39]): passes for $q \in \{61, 151, 211, 307\}$ with variance ratio $R_{\text{var}} \leq 1$ relative to the control noise floor.

7.4 Open hypotheses

The following items remain open within the sub-programme.

1. $[\text{H-color}]_{\text{pointwise}}$: proved analytically in O31 (Proposition 4.23, [40]): each O12 fingerprint vector carries a single frequency $B = c_1 b_1 + c_2 b_2 + c_3 b_3$, and the orbit dilation $B \mapsto \omega B$ is a bijection of $\mathbb{Z}/q\mathbb{Z}$. Together with the three prior levels, [H-color] is fully established.

2. *Asymptotic stabilisation of* δ_∞: the limit $\delta_\infty = \lim_{q\to\infty} \delta_{\text{pair}}(q)$ has not been computed analytically. The correct asymptotic variable is $n_1(q)/q$, not q itself.

3. *Extension to* $q = 401$: the campaign at $q \in \{29, 61, 101, 151, 211, 307\}$ is complete; $q = 401$ is pending.

8 Open Deliverables

Two open deliverables define the boundary of the sub-programme as of this note.

Closed: $[\text{H-color}]_{\text{pointwise}}$**.** All four levels of [H-color] are now analytically established. (i) $[\text{H-color}]_{\text{rank}}$: sector rank equality $r_c(n) = r_{\omega c}(n) = r_{\omega^2 c}(n)$ proved unconditionally in O31 [40] via intra-coset Vandermonde symmetry. (ii) $[\text{H-color}]_{\text{avg}}$: proved in O32 (Proposition 1) to $O(q^{-1})$. (iii) $[\text{H-color}]_{\text{eff}}$: proved in O32 in the $q \to \infty$ limit. (iv) $[\text{H-color}]_{\text{pointwise}}$: proved in O31 (Proposition 4.23, [40]) via the single-frequency fingerprint structure. The SU(3) identification on the standard graph is therefore fully unconditional.

Open deliverable 2: Numerical campaign at $q = 401$**.** The campaign at $q \in \{29, 61, 101, 151, 211, 307\}$ is complete. $q = 307$ confirms [H-color] ($R_{\text{var}} = 1.17 \times 10^{-3}$, all 4/4 triplets, O32 [39]). The extension to $q = 401$ is the sole remaining numerical task: it constrains the rate of convergence of $n_1(q)/q$ and provides the fifth data point for the $R_{\text{var}}(q) \propto q^{-1}$ scaling prediction.

All other open problems of earlier papers (e.g., the rigorous continuum limit of the state law, the exact form of Φ on LPS graphs) are sub-problems within already resolved phases and do not block the downstream branches.

9 Conclusion

The spectral admissibility sub-programme provides the core computational content of the Cosmochrony corpus. Starting from the Born–Infeld bounded-flux constraint on the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, it derives the cascade exponent $\beta^* \approx 0.126$ unconditionally, identifies the spin-$\frac{1}{2}$ admissible sector, and establishes the admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$ as a structural output.

The sub-programme is analytically closed for the SU(2) sector. It is closed numerically for $q \in \{29, 61, 101, 151, 211, 307\}$. The extension to SU(3) is established unconditionally for the colour-adapted graph and at all four levels on the standard graph (O31–O32, [40, 39]): $[\text{H-color}]_{\text{pointwise}}$ is proved analytically in O31 (Proposition 4.23), completing the SU(3) identification without residual conditionality.

The outputs of this sub-programme are used throughout Branch III: by the quantum sector (Q1–Q3), by the geometric emergence sub-programme (Q5a–Q11), by the gauge structure sub-programme (Q6a, O31–O32), and by the fermionic matter papers (Q14). Reading the O-series without this map is possible but significantly more difficult; this note is intended to make the logical structure transparent before the technical machinery is engaged.

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